CSE211:COMPUTER ORGANIZATION AND DESIGN

L:3 T:1 P:0 Credits:4

Course Outcomes: Through this course students should be able to

CO1 :: define the basics of organization and architecture of a digital computer

CO2 :: explain the concept of Instruction sets, microprocessor design and addressing modes

CO3 :: examine the input-output unit and its processing in a digital computer

CO4 :: analyze Verilog HDL, in relation of Register Transfer to model, simulate, and verify digital systems

CO5 :: articulate the performance of various computer memory systems and analyze parallel processing

CO6 :: analyze modern computing architectures, including GPU architecture and understand latest trends in computer architecture

Unit I

Computer Organization : instruction codes, computer registers, common bus system, computer instructions, timing and control, instruction cycle, memory reference instructions, input-output and interrupt, lmc(simulation software)

Unit II

Central Processing Unit : General Register Organization, Stack Organization, Addressing Modes, Reduced instruction set computer, Complex instruction set computer, instruction formats, Data Transfer Schemes, Program Control and Interrupts, Processor status word

Unit III

Input-Output Organization : Peripheral Devices, Priority interrupt, Input output interface, Direct memory access transfer, Input/Output processor., modes of data transfer, UART

Unit IV

Introduction to HDLs in Digital Systems : History, Evolution, Typical HDL-based Design Flow (Design, Simulation, Synthesis, Verification), Comparison with Software Programming Languages, Verilog, Register Transfer and Micro Operation, Register Transfer Language, Bus and Memory Transfer, Shift Register, Micro operations (Arithmetic, logic, shift), Addition, Subtraction, Multiplication Algorithm, Booth Multiplication

Unit V

Memory Unit : Memory hierarchy, Cache memory, Virtual memory, main memory, auxiliary memory, Writing into Cache concept, Mapping Techniques, Introduction to Parallel Processing, Pipelining, Characteristics of Multiprocessors, Interconnection Structures, Parallel Processing

Unit VI

Introduction to GPU Architecture : Nvidia Case Study, Introduction to Supercomputer, Introduction to Qubits and Quantum Computing, Latest Technology and Trends in Computer Architecture, Next Generation Processors Architecture, Microarchitecture, Latest Processor for Smartphone or Tablet and Desktop

Text Books:

1. COMPUTER SYSTEM ARCHITECTURE by M. MORRIS MANO, RAJIB MALL, PEARSON

References:

1. DIGITAL SYSTEMS DESIGN USING VERILOG by CHARLES ROTH, LIZY K. JOHN, BYEONG KIL LEE, Cengage India Private Limited

2. COMPUTER ORGANIZATION AND ARCHITECTURE by WILLIAM STALLINGS, Pearson Education India